

Generate the data of size 20 for vector $X' = [X_1, X_2, X_3]$ using multivariate normal distribution with mean vector

$\mu = [160, 60, 30]$ and covariance matrix $\Sigma = \begin{bmatrix} 4 & 3 & 2 \\ 3 & 6 & 5 \\ 2 & 5 & 9 \end{bmatrix}$. Also, calculate the sample mean vector, covariance matrix

and correlation matrix of vector X .

Creating a Matrix Mu and Sigma

```
set.seed(123)
mu<-matrix(c(160,60,30),nrow=3,byrow=TRUE)

sigma<-matrix(c(4,3,2,3,6,5,2,5,9),3,3)
```

Generating a random variable size=20 from multivariate normal distribution

```
library(MASS)
z<-mvrnorm(20,mu,sigma)
z

##           [,1]      [,2]      [,3]
## [1,] 161.7251 62.51052 30.05433
## [2,] 160.4480 60.82456 30.30381
## [3,] 158.6796 58.30667 24.06035
## [4,] 162.4512 58.43617 29.80650
## [5,] 161.5670 59.06951 29.39477
## [6,] 159.5273 58.25803 22.93109
## [7,] 157.9651 58.80103 29.53540
## [8,] 161.0059 63.09269 33.47193
## [9,] 162.9995 61.56194 30.88721
## [10,] 158.7024 60.25951 32.62390
## [11,] 158.0712 56.81421 27.22075
## [12,] 159.9603 59.41874 28.66355
## [13,] 158.2060 58.58679 29.90427
## [14,] 159.5610 58.01965 31.22359
## [15,] 159.3540 60.89892 32.37853
## [16,] 157.8833 54.31875 26.45835
## [17,] 157.5317 59.88756 28.67012
## [18,] 162.8972 63.86841 35.55719
## [19,] 159.6636 58.54166 27.77117
## [20,] 161.2691 61.09660 30.94757
```

Sample Mean Vector

```
sample_mean_vector<-colMeans(z)
sample_mean_vector

## [1] 159.97342 59.62860 29.59322
```

Variance Covariance Matrix

```
var_cov_matrix<-cov(z)
var_cov_matrix

##           [,1]      [,2]      [,3]
## [1,] 3.035117 2.450235 2.411195
## [2,] 2.450235 4.986501 4.710670
## [3,] 2.411195 4.710670 8.953396
```

Correlation of Sample Covariance Matrix

```
diag_sqrt_var<-sqrt(diag(diag(var_cov_matrix),nrow = 3))

inverse_daig_matrix<-solve(diag_sqrt_var)

corr_result<-inverse_daig_matrix%%var_cov_matrix%%inverse_daig_matrix
corr_result

##           [,1]      [,2]      [,3]
## [1,] 1.0000000 0.6298283 0.4625415
## [2,] 0.6298283 1.0000000 0.7050028
## [3,] 0.4625415 0.7050028 1.0000000
```

Verification of The Manually Calculated Result

```
cor_result<-cor(z)
cor_result

##           [,1]      [,2]      [,3]
## [1,] 1.0000000 0.6298283 0.4625415
## [2,] 0.6298283 1.0000000 0.7050028
## [3,] 0.4625415 0.7050028 1.0000000
```